

Course Registration System Backlog

Stakeholders & Users:

Students: Search for courses, register for course sessions, view their enrolled sessions, and receive registration confirmations.

Instructors: View the sessions they are assigned to teach and the rosters for those sessions.

Administrators: Create and update course sessions and manage the overall system.

Registrar/University Administration: Needs accurate enrollment information, including the daily enrollment report.

University Catalog System (external system): Supplies course data to the system nightly.

User Story 1: Student Login

User Story:

As a student, I want to log into the course registration system so that I can access my student account and registration features.

Description:

The system should allow students to log into the course registration system. After successfully logging in, the student can access the features available to them.

Priority: High

Source:

POST /students/login Log a student in.

Priority Justification:

Login is a high priority because students need to access the system before they can use student features such as searching for courses and registering for sessions.

Acceptance Criterion:

Given that a student has valid login information, when the student submits their login information, then the system allows the student to access the course registration system.

User Story 2: Search for Courses

User Story:

As a student, I want to search the course catalog so that I can find courses that I am interested in taking.

Description:

The system should allow students to search the course catalog and view matching course information.

Priority: High

Source:

GET /courses Search the course catalog.

Priority Justification:

Course search is important because students need to find courses before deciding which sessions they want to register for. It is also one of the main functions included in the team's Course Registration System.

Acceptance Criterion:

Given that a student is using the course registration system, when the student searches the course catalog, then the system displays the matching course information.

User Story 3: Register for an Available Session

User Story:

As a student, I want to register for an available course session so that I can enroll in the session I want to take.

Description:

The system should allow students to select a session that is open for registration. Before completing the registration, the system checks whether the session still has available seats. If space is available, the student's registration is recorded.

Priority: High

Source:

GET /sessions/available Show sessions open for registration.

POST /sessions/{id}/register Register a student for a session. Checks if there is room first.

Priority Justification:

Registration is a high priority because it is a primary function of the Course Registration System. The system also needs to check session availability before creating the student's registration.

Acceptance Criterion:

Given that a student selects a session that is open for registration, when the student registers for the session, then the system checks whether there is an available seat and records the registration if space is available. If the session is full, the system does not complete the registration.

User Story 4: View Course Details

User Story:

As a student, I want to be able to view a course's details and info so that I can determine if it fits my schedule and degree requirements.

Description:

Students need to view important information about courses before they can decide about registering for the courses.

Priority: Medium

Priority Justification:

While viewing course details is useful, other functionalities like core course registration and enrollment are more critical.

Acceptance Criterion:

The system should display the course name, course number, number of credits, prerequisites of the course, max enrollment, and the student can view course details before registering.

Source: Course Search screen from the UI/UX Assignment.

User Story 5: Nightly Catalog Update

User Story:

As a registrar, I want course data to update nightly from the university catalog so that students can always see up to date course information.

Description:

The Course Registration System must receive current course information from the university on a nightly basis. The catalog data is read-only within the CRS (Course Registration System).

Priority: Medium

Priority Justification: Keeping course info current is important for accurate registration for students and university staff, but not as critical as core registration functions.

Acceptance Criterion:

The system reads course data from the university catalog nightly. Updated catalog information is available in the CRS after each nightly update. The system can read the catalog data but cannot modify the catalog data. Students can see the updated info after each nightly update, and users cannot edit the catalog information through the CRS.

Source:

Function Point Assignment - nightly read-only course data from the university catalog.

User Story 6: View Enrolled Sessions

User Story:

As a student, I want to view my currently enrolled course sessions so that I know my class schedule when needed.

Description:

The system should allow a student to view the course sessions they are currently enrolled in, allowing students to review their current class schedule after they register for courses when needed.

Priority: Medium

Priority Justification:

Viewing enrolled course sessions provides important value to students because it allows them to check and verify their schedules. It however depends on core enrollment functionality being available, so it is less critical than primary registration functionality.

Acceptance Criterion:

The system retrieves the enrolled course sessions of the student, allowing them to view all of their course sessions. The sessions display relevant course and session information. The displayed sessions correspond to the student's enrollment records only. The system uses GET /students/ {id} /enrollments endpoint to retrieve the student's enrolled sessions.

Source:

Architecture Document - GET /students/{id}/enrollments endpoint.

User Story 7: Prerequisite Check**User Story:**

As a student, I want the system to prevent me from registering for a course when I have not completed its prerequisites so that I only enroll in courses I am qualified to take.

Description:

Before completing a course registration, the system should verify that the student has satisfied all prerequisite requirements associated with the selected course. If the prerequisites have not been met, the registration is denied and the student is informed of the missing prerequisite(s).

Priority: Medium**Source:**

Prerequisite table (Database Assignment) and prerequisites displayed on the Course Search screen (UI/UX Assignment).

Priority Justification:

This story is a medium priority because enforcing prerequisites is an important academic business rule that ensures students register for appropriate courses. However, it depends on the core registration functionality and is less critical than the ability to search for and register for courses.

Acceptance Criterion:

Given that a student selects a course session for registration, when the system checks the course prerequisites, then the system allows registration only if all prerequisites have been satisfied. If one or more prerequisites have not been met, the system denies the registration and displays the missing prerequisite requirement(s).

User Story 8: Administrator Manages Sessions

User Story:

As an administrator, I want to create and update course sessions so that students have accurate sessions available for registration.

Description:

The system should allow authorized administrators to create new course sessions and update existing session information. Administrators can manage session details such as section number, meeting days, meeting times, instructor assignment, location, and maximum enrollment. These changes ensure that students can register for accurate and current course sessions.

Priority: Medium

Source:

Administrator role and Authentication & Authorization section (Architecture Document).

Priority Justification:

This story is a medium priority because administrators must be able to maintain accurate session information for the registration system to function correctly. However, it is not as critical as the core student functions of logging in, searching for courses, and registering for sessions.

Acceptance Criterion:

Given that an authorized administrator is logged into the system, when the administrator creates a new course session or updates an existing session, then the system saves the session information and makes the updated session details available to students during course registration.

User Story 9: Instructor Views Assigned Sessions

User Story:

As an instructor, I want to log into the course registration system and view my assigned course sessions so that I know which courses I am teaching and when they are scheduled.

Description:

The system should allow instructors to log in using their instructor credentials and access their teaching schedule. After logging in, instructors can view all course sessions assigned to them, including course information, section number, meeting days, meeting times, and location.

Priority: Medium

Source:

POST /instructors/login and GET /instructors/{id}/schedule endpoints (Architecture Document).

Priority Justification:

This story is a medium priority because instructors need access to their teaching schedules to perform their responsibilities. However, it is less critical than the core student registration functions, since students can still search for and register for courses without instructor schedule access.

Acceptance Criterion:

Given that an instructor has valid login credentials, when the instructor logs into the system, then the system authenticates the instructor and displays all course sessions assigned to that instructor. The schedule includes the course name, section number, meeting days, meeting times, and location. Only the logged-in instructor's assigned sessions are displayed.

User Story 10: Daily Enrollment Report (Business rule/operational constraint)

User Story:

As a registrar, I want to receive a daily enrollment report so that I can track registration activity across all sessions.

Description:

Once a day, the system emails the registrar a report summarizing enrollment for every session. The report lists each session's course number, section, number of enrolled students, and maximum enrollment, so the registrar can see which sessions are filling up or already full.

Source:

Function Point Assignment requirements: a daily enrollment report sent to the registrar. The team chose email as the delivery method.

Priority: Low

Priority Justification:

The report is a low priority because students can search for and register for sessions without it. It supports the registrar's oversight of registration but is not part of the core registration process.

Acceptance Criterion:

Given that it is the scheduled time for the daily report, when the system generates the report, then the registrar receives an email containing a report that lists every session with its enrolled count and maximum enrollment.

User Story 11: Registration Confirmation Email

User Story:

As a student, I want to receive a confirmation email after registering for a session so that I have a record of my enrollment.

Description:

After a student successfully registers for a session, the system sends the student an email confirming the registration. The email includes the course, section, days, time, and location of the session. No email is sent if the registration is not completed.

Source:

Function Point Assignment requirements: a confirmation email to students after successful registration.

Priority: Low

Priority Justification:

The confirmation email is a low priority because registration still works without it, and the system already displays an on-screen confirmation message after a successful registration. The email gives students an added record of their enrollment.

Acceptance Criterion:

Given that a student has successfully registered for a session, when the registration is recorded, then the system sends the student an email containing the session's course, section, days, time, and location.

User Story 12: Session Roster with Restricted Access (Business rule/operational constraint)

User Story:

As an instructor, I want to view the roster for each session I teach so that I can see which students are enrolled.

Description:

The system allows an instructor to view the roster for the sessions assigned to them. The roster shows each student's ID, name, and enrollment status, along with the number of students enrolled compared to the session's maximum enrollment. Because rosters contain student information, access is restricted to the instructor assigned to the session and to administrators. Students and other instructors cannot view the roster.

Source:

GET /sessions/{id}/roster endpoint and the Instructor role in the Authentication & Authorization section (Architecture Document); Session Enrollment screen, which is accessible to authorized users only (UI/UX Assignment).

Priority: Medium

Priority Justification:

The roster is a medium priority because instructors need to know who is enrolled in their sessions, and protecting student information is required for the system to be used responsibly. However, students can still search for and register for sessions without this feature.

Acceptance Criteria:

Given that an instructor is assigned to a session, when the instructor views that session's roster, then the system displays each enrolled student's ID, name, and status, along with the enrolled count and maximum enrollment.

Given that a user is not the session's assigned instructor or an administrator, when the user tries to view the roster, then the system denies access.